
Software Requirements Specification

for

Local Tourist Day-Visit Planner and Information System (Kandy, Sri Lanka)

Version 1.1

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Revision History

Name	Date	Reason For Changes	Version
Krishalani Thiyagaraja (E2410433)	August 21, 2026	Initial SRS drafted based on the approved Project Proposal, following the IEEE/Wiegers SRS template required for Assignments 2 and 3.	Version 1.0
Krishalani Thiyagaraja (E2410433)	September 6, 2026	Incorporated Week 7 peer-review feedback: added visual analysis models (System Architecture diagram, Use Case Diagram, and Conceptual Data Model ER diagram) to Appendix B.	Version 1.1

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete set of functional and non-functional requirements for the **Local Tourist Day-Visit Planner and Information System**, a web-based application for tourists visiting Kandy, Sri Lanka. This document covers Version 1.0 of the system, corresponding to the full scope described in the approved Project Proposal (*ITE2953_ProjectProposal_E2410433*, submitted for module ITE2953 - Programming Group Project).

This SRS describes the complete system: the tourist-facing attraction catalogue, search and filtering, attraction detail and map views, the one-day itinerary planner, and the administrator module used to maintain attraction data. It does not describe any subsystem partially; all modules identified in the Project Proposal are covered in full.

1.2 Document Conventions

- Every mandatory, binding requirement is stated using the word “**shall**”, consistent with IEEE SRS convention (e.g., “The system shall...”). Where the source Project Proposal used softer language such as “should” or “users shall be able to,” these statements have been normalized to “shall” here so that every requirement in this document carries the same binding weight and can be tested consistently.
- Functional requirements are uniquely identified with the prefix **FR-** followed by a two-digit sequence number (e.g., FR-01) and are grouped under the System Feature that they support (Section 4).
- Non-functional requirements are uniquely identified with the prefix **NFR-** followed by a two-digit sequence number (e.g., NFR-01) (Section 5).
- User/stakeholder requirements are uniquely identified with the prefix **UR-** (Section 2.7.1) and are traced forward to the functional requirements that satisfy them (Section 6.2, Requirements Traceability Matrix).
- Risks identified in the Project Proposal are referenced using the prefix **R-** (e.g., R1) where relevant to a requirement’s rationale.
- All requirements in this document are assumed to carry equal (mandatory) priority unless a specific priority (High/Medium/Low) is stated for the System Feature that contains them, in Section 4.
- Text shown in *italics* within a requirement is used only for clarifying examples and is not itself a binding part of the requirement.
- Items that could not be resolved from the Project Proposal and require a decision during the System Design phase are explicitly marked **[TBD]** and collected in Appendix C, rather than being assumed.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for:

- **The lecturer/examiner** assesses this document against the Assignment 2 and 3 SRS requirements and the IEEE SRS structure.
- **The developer (the author)**, who will use this document as the authoritative basis for system design (database design, UI wireframes, and the Web API contract) and implementation in the subsequent project phases.
- **Peer reviewers**, who will evaluate the SRS as part of the module's peer-evaluation activity (Week 7 of the module schedule).
- **Testers** (including the developer acting in a testing capacity), who will use the functional and non-functional requirements in Sections 4 and 5 to design test cases and verify the completed system.

Readers unfamiliar with the project should read Sections 1 and 2 first for context, then Section 4 (System Features) for the core functional scope. Readers who need the full behavioral detail should also read Section 3 (External Interfaces) and Section 5 (Non-Functional Requirements). Section 6 and the appendices provide supporting material, including the traceability matrix, glossary, and outstanding items.

1.4 Product Scope

The Local Tourist Day-Visit Planner and Information System is a web-based application that centralizes information about tourist attractions within approximately 25 km of Kandy, Sri Lanka, and allows tourists to search, filter, and view detailed attraction information, view attraction locations on an interactive map, and assemble a personalized one-day visit plan. A separate administrator module allows an authorized administrator to keep the attraction catalogue accurate and up to date.

The system addresses a specific, documented business problem: tourist information about the Kandy area is scattered across multiple websites and social media platforms, making it time-consuming for tourists to compare destinations and plan an efficient one-day visit (Project Proposal, Section 1.1). By consolidating verified attraction information, search/filter capability, and itinerary planning into a single platform, the system is intended to reduce the time tourists spend gathering information and to support more efficient day-trip planning, while also promoting local tourism in the Kandy area (Project Proposal, Sections 1.2 and 1.14).

In scope (from Project Proposal, Section 1.4.1):

- A catalogue of at least 15 tourist attractions located within approximately 25 km of Kandy, categorized as Religious & Heritage, Nature, Adventure, Museum, Viewpoint, Recreation, or Eco Tourism.
- Search of attractions by place name and filtering by category.
- Detailed attraction information: description, opening hours, travel tips, approximate distance, images, and map location.
- A personalized one-day visit plan that tourists can build by selecting preferred attractions.
- An administrator module to add, edit, and delete attraction records.
- A responsive web application is usable on desktop and mobile browsers.

Explicitly out of scope (from Project Proposal, Section 1.4.2) these exclusions are binding constraints on this SRS and no requirement in this document contradicts them:

- Online hotel booking services.
- Online transportation or vehicle reservation facilities.
- Online ticket booking and payment processing.
- Tourist user registration and online travel package management.
- Attractions are located beyond approximately 25 km of Kandy.

1.5 References

1. K. Thiagaraja, “ITE2953 - Programming Group Project: Project Proposal — Local Tourist Day-Visit Planner and Information System (Kandy, Sri Lanka),” University of Moratuwa, Faculty of Information Technology, 2026. (Source proposal for this SRS.)
2. K. E. Wieggers, “Software Requirements Specification Template,” Copyright 1999, used under license as the compulsory IEEE-style SRS template for this assignment (*srs_template-ieee.pdf*).
3. S. Gunawardena, “Requirement Analysis — Mastering the Software Requirements Specification (SRS),” ITE2953 - Programming Group Project lecture notes, University of Moratuwa (*Creating an SRS Document.pdf*).
4. Sri Lanka Tourism Development Authority, “Official Sri Lanka Tourism Website.” [Online]. Available: <https://www.sltda.gov.lk>
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2. Overall Description

2.1 Product Perspective

The Local Tourist Day-Visit Planner and Information System is a **new, self-contained product**; it is not a replacement for, or a follow-on to, an existing system. It was proposed in response to a documented gap in the local tourism sector: no single, trusted, centralized source of attraction information exists for tourists planning a one-day visit to Kandy (Project Proposal, Section 1.1).

The system follows a standard **three-tier web architecture**, summarized below (adapted from Project Proposal, Section 1.7, Figure 1):

Layer	Component	Technology	Responsibility
Presentation	Angular Single Page Application (SPA)	Angular, TypeScript, HTML5, CSS3	Renders the user interface and handles user interaction (search, filter, itinerary builder, administrator forms).
Application / API	ASP.NET Core Web API	C#, ASP.NET Core, RESTful API	Implements business logic, exposes RESTful endpoints, and enforces authentication and authorization.
Data	SQL Server Database	SQL Server, Entity Framework Core	Stores tourist attraction records, categories, and administrator accounts.
External Service	Google Maps Integration	Google Maps API	Provides interactive maps, distances, and location information for each attraction.

The Angular frontend communicates with the ASP.NET Core Web API over HTTP using JSON, consistent with the standard client-server request/response model. The system integrates with one external third-party service, the Google Maps API, for map display and location information; this is the system's only external interface (see Section 3.3 and 3.4).

2.2 Product Functions

The major functions the system must provide are summarized below; detailed functional requirements are provided in Section 4.

- Display a catalogue of tourist attractions located within approximately 25 km of Kandy.
- Allow a tourist to search attractions by place name and filter them by category.
- Display detailed information for a selected attraction (description, opening hours, travel tips, approximate distance, images, map location).
- Display each attraction's location on an interactive map.
- Allow tourists to build, view, and modify a personalized one-day visit plan from selected attractions.

- Provide secure administrator login.
- Allow an authenticated administrator to add, update, and delete tourist attraction records.
- Provide a responsive interface that is usable on desktop and mobile web browsers.

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise	Frequency of Use	Importance
Tourist User	Local residents, domestic travelers, or international visitors who wish to explore attractions within 25 km of Kandy and plan a one-day visit. May access the system on desktop or mobile devices. No account or registration is required.	Low to moderate; assumed to have basic web-browsing skills only.	Typically, occasional/short sessions concentrated around trip-planning (maybe a single visit before a trip).	Primary — the system's core value is delivered to this class.
System Administrator	The individual(s) responsible for maintaining the accuracy of the attraction catalogue: adding new attractions, correcting or updating existing records, and removing outdated or incorrect ones. Authenticates with a username and password.	Moderate; comfortable with structured data-entry forms.	Periodic (as new attractions are identified or existing data needs correction), lower frequency than tourist use.	Secondly, critical — data quality for all Tourist Users depends on this class.

2.4 Operating Environment

- **Client side:** Any device with a modern web browser and an active internet connection (Project Proposal, Section 1.15, Assumptions). The Angular SPA must render and operate correctly on desktop and mobile screen sizes (Project Proposal, Section 1.4.1, and NFR-08 in Section 5.4 of this document).
- **Target browsers:** Google Chrome, Microsoft Edge, and Mozilla Firefox (current major versions), consistent with the Compatibility non-functional requirement in the Project Proposal (Section 1.11, NFR8). Exact minimum supported browser version numbers are [TBD] (see Appendix C).
- **Server side:** ASP.NET Core Web API runtime (C#), which requires a compatible .NET hosting environment.
- **Data tier:** Microsoft SQL Server, accessed via Entity Framework Core.

- **Hosting infrastructure** (cloud provider, server specification, or on-premises deployment target) is not specified in the Project Proposal and is recorded as [TBD] in Appendix C; it is expected to be resolved during the System Design phase (Project Proposal, Section 1.12).

2.5 Design and Implementation Constraints

- **Fixed technology stack:** Development is constrained to Angular (frontend), ASP.NET Core Web API with C# (backend/application layer), SQL Server (data layer), and Entity Framework Core as the ORM, as specified in the Project Proposal, Section 1.9. No alternative frameworks or languages may be substituted without a documented change to the proposal.
- **Geographic scope constraint:** Only attractions located within approximately 25 km of Kandy may be included in the catalogue (Project Proposal, Sections 1.4.1 and 1.4.2).
- **Third-party API dependency:** Map, distance, and location features are constrained by the capabilities, quotas, and terms of use of the Google Maps API (Project Proposal, Section 1.9 and Risk R5, Section 1.13).
- **Single-developer delivery:** Although consultation for requirements elicitation involved four peer stakeholders (Section 2.7.1), this is an **individual** project; design and implementation are carried out by one developer (Krishalani Thiyagaraja, E2410433), which constrains the achievable scope within the fixed academic timeline.
- **Fixed academic timeline:** The project must be delivered within the 14-week ITE2953 module schedule, with solution/database design in Weeks 7–8 and implementation in Weeks 9–11 (Project Proposal, Sections 1.12–1.13 and Appendix B).
- **Version control:** Implementation must use Git and GitHub, per the Project Proposal, Section 1.9.
- **Development tooling:** Visual Studio Code is the specified development tool (Project Proposal, Section 1.9); this is a tooling preference and does not constrain the system’s runtime behavior.

2.6 User Documentation

The Project Proposal does not specify a formal set of user-documentation deliverables (such as a user manual, online help system, or tutorial). Given the system’s stated usability goal — a “user-friendly and responsive web application” (Project Proposal, Section 1.3) — the interface is expected to be self-explanatory for both Tourist Users and the Administrator without a separate manual, supported by on-screen labels, placeholder text, and validation/error messages defined in Sections 3.1 and 4. Whether a standalone user guide or administrator guide will be produced as a project deliverable is [TBD] (see Appendix C).

2.7 Assumptions and Dependencies

Assumptions (from Project Proposal, Section 1.15):

- Tourist attraction data (descriptions, opening hours, distances) will be sourced from publicly available and official information and verified before publication (see also Risk R1, Section 1.13 of the proposal, which recommends cross-verification against sources

such as the Sri Lanka Tourism Development Authority, the Department of Wildlife Conservation, and the Department of Archaeology).

- Users will have access to a modern web browser and an active internet connection when using the system.
- The developer and peer group members will remain available for stakeholder consultation and testing throughout the project timeline.

Dependencies:

- The system depends on the continued availability and correct functioning of the **Google Maps API** for all location, distance, and map-display features (FR-08). Any change to the Google Maps API's terms, pricing, or usage quotas could affect this dependency (Risk R5, Project Proposal Section 1.13).
- The accuracy and completeness of the attraction catalogue depend on the System Administrator maintaining the data over time (Project Proposal, Section 1.4.2, "Tourist information depends on the data maintained by the system administrator and may require periodic updates").
- The application layer depends on Entity Framework Core correctly mapping between the C# domain model and the SQL Server schema.

2.7.1 User/Stakeholder Requirements

The requirements in this SRS were derived from the Project Objectives (Project Proposal, Section 1.3) and from structured stakeholder consultation with four peer group members acting as representative stakeholders (Project Proposal, Section 1.6 and Appendix A). The table below states each user/stakeholder requirement in "As a ... I want ... so that ..." form and identifies its source. These requirements are traced forward to the functional requirements that satisfy them in the Requirements Traceability Matrix (Section 6.2).

ID	User/Stakeholder Requirement	Source
UR-01	As a tourist, I want to view a catalogue of attractions within 25 km of Kandy with essential details (name, description, photos, opening hours, travel information), so that I do not have to gather information from multiple scattered websites.	Project Objective 1; Stakeholder James JBK (E2410439)
UR-02	As a tourist, I want to search for an attraction by name and filter the catalogue by category, so that I can quickly narrow down attractions that match my interests.	Project Objective 2; Stakeholder Werasekara SNDRWMMGJCB (E2410436)
UR-03	As a tourist, I want to see each attraction's location, distance, and travel information, so that I can judge how practical it is to include my day trip.	Project Objective 3; Stakeholder Seema AFZ (E2410442)

ID	User/Stakeholder Requirement	Source
UR-04	As a tourist, I want to select multiple attractions and build a personalized one-day visit plan, so that I can organize an efficient day trip without manual notetaking.	Project Objective 4; Stakeholders Seema AFZ (E2410442) and Werasekara SNDARWMMGJCB (E2410436)
UR-05	As a system administrator, I want to securely add, update, and delete attraction records, so that the information tourists rely on stays accurate and up to date.	Project Objective 5; Stakeholder Madhushan S (E2410438)
UR-06	As a tourist, I want the application to work well on both my phone and my computer, so that I can plan my trip conveniently regardless of the device I have available.	Project Objective 6

3. External Interface Requirements

3.1 User Interfaces

The system provides a single web-based user interface (the Angular SPA), presented differently depending on the authenticated role:

- **Attraction list / home view:** Displays the catalogue of attractions with a search field and category filter controls (supports FR-01 to FR-06).
- **Attraction detail view:** Displays full details for a selected attraction, including an embedded interactive map (supports FR-07 to FR-09).
- **Itinerary planner view:** Displays the tourist's currently selected attractions for their one-day visit plan, with controls to add/remove attractions (supports FR-10 to FR-13).
- **Administrator login view:** A form for entering administrator credentials (supports FR-14, FR-15).
- **Administrator management view:** Forms and list views for adding, editing, and deleting attraction records, restricted to authenticated administrators (supports FR-16 to FR-21).

General UI conventions:

- The layout must be responsive, adapting to both desktop and mobile browser widths (see NFR-08, Section 5.4).
- Every user-facing error condition defined in Section 4 (e.g., no search results, invalid login, missing mandatory field) must be communicated to the user through a visible, on-screen message; the system shall not fail silently.
- Detailed screen layouts, wireframes, and visual style are design-phase artifacts and are outside the scope of this SRS (Project Proposal, Section 1.12, "System Design" phase); this document specifies only the logical interface requirements needed to satisfy the functional requirements in Section 4.

3.2 Hardware Interfaces

The system has no direct hardware interfaces. It is a web application consumed through a standard web browser; the only implied hardware requirement is that the client device (desktop, laptop, tablet, or smartphone) must be capable of running one of the target browsers listed in Section 2.4 and of rendering an interactive web map. Server-side hardware/hosting specifications are **[TBD]** (Appendix C).

3.3 Software Interfaces

Interface	Name / Version	Purpose
Frontend framework	Angular	Implements the presentation layer (SPA) consumed by the tourist and administrator.
Backend framework	ASP.NET Core Web API (C#)	Implements business logic and exposes RESTful endpoints consumed by the Angular frontend.

Interface	Name / Version	Purpose
ORM / Data access	Entity Framework Core	Maps the C# domain model to the SQL Server database; used exclusively by the ASP.NET Core Web API layer.
Database	Microsoft SQL Server	Stores attraction records, categories, and administrator accounts (Section 6.1).
External mapping service	Google Maps API	Provides map rendering, attraction location markers, and distance/location data consumed by the attraction detail view (FR-08).
Client runtime	Modern web browser (Chrome, Edge, Firefox)	Executes the Angular SPA and renders the embedded Google Maps component.

3.4 Communications Interfaces

- The Angular frontend and the ASP.NET Core Web API communicate over **HTTP(S)** using **RESTful endpoints**, exchanging data in **JSON** format (Project Proposal, Section 1.7).
- The frontend communicates with Google **Maps API** over HTTPS to retrieve map tiles, markers, and location data for the attraction detail view (FR-08).
- Administrator authentication requests (FR-14) are transmitted from the Angular frontend to the ASP.NET Core Web API; the credential-transport security mechanism (e.g., token-based authentication such as ASP.NET Core Identity, referenced in NFR-04) is a design-phase decision. Whether the deployed system enforces TLS/HTTPS end-to-end is not explicitly stated in the Project Proposal and is recorded as **[TBD]** in Appendix C, though it is recommended as good practice given that administrator credentials are transmitted.

4. System Features

This section itemizes the detailed functional requirements of the system, organized into five System Features that together implement the ten high-level functional requirements identified in the Project Proposal (Section 1.10, FR1–FR10). Each proposal-level requirement has been decomposed into atomic, individually testable statements, and reasonable, directly implied error-handling behaviour (e.g., “no results found,” “invalid login,” “confirm deletion”) has been made explicit, consistent with the guidance that an SRS must state how the system responds to anticipated error conditions. No functionality beyond what is described or directly implied by the Project Proposal has been introduced.

4.1 Attraction Catalogue, Search, and Category Filtering

4.1.1 Description and Priority

Allow a tourist to browse the full catalogue of attractions and narrow it down by place name or category. This is the entry point of the tourist experience and is rated **High** priority, since every other tourist-facing feature (detail view, map, itinerary) depends on the tourist first finding an attraction through this feature.

4.1.2 Stimulus/Response Sequences

- Tourists open the application → system displaying the full catalogue of attractions.
- Tourist enters text in the search field → system displays only attractions whose name matches the entered text.
- Tourist selects one or more category filters → system displays only attractions belonging to the selected category or categories.
- Tourists combine a search term with a category filter → system displays only attractions that satisfy both conditions.

4.1.3 Functional Requirements

- **FR-01:** The system shall display a list of tourist attractions located within approximately 25 km of Kandy, showing at minimum the name, category, and a representative image for each attraction.
- **FR-02:** The system shall allow the tourist to enter a place name into a search field.
- **FR-03:** The system shall display the subset of attractions whose name matches the tourist’s entered search text.
- **FR-04:** The system shall display a “no attractions found” message when a search query does not match any attraction in the catalogue.
- **FR-05:** The system shall allow the tourists to filter the displayed attractions by one or more of the following categories: Religious & Heritage, Nature, Adventure, Museum, Viewpoint, Recreation, and Eco Tourism.
- **FR-06:** The system shall allow the tourists to apply a search term and a category filter together, displaying only attractions that satisfy both criteria.

4.2 Attraction Detail and Interactive Map View

4.2.1 Description and Priority

Allows a tourist to view complete information about a specific attraction, including its location on an interactive map, in order to decide whether to include it in their visit plan. Rated **High** priority, as detailed and location information is central to the system's stated purpose of supporting informed travel decisions (Project Objective 3).

4.2.2 Stimulus/Response Sequences

- Tourist selects an attraction from the catalogue list → system displays the attraction's full detail view.
- Tourist views the detail page → system displays the attraction's location on an interactive map.
- Tourists navigate back from the detail view → system returns to the catalogue list, preserving any previously applied search or filter.

4.2.3 Functional Requirements

- **FR-07:** The system shall display detailed information for a selected attraction, including its name, description, category, opening hours, travel tips, images, and approximate distance from Kandy.
- **FR-08:** The system shall display the geographic location of the selected attraction on an interactive map using the Google Maps API.
- **FR-09:** The system shall return the tourist to the attraction catalogue list, with any previously applied search text or category filter still in effect, when the tourist navigates back from the attraction detail view.

4.3 One-Day Visit Itinerary Planner

4.3.1 Description and Priority

Allow a tourist to assemble a personalized set of attractions into a one-day visit plan. This is the feature that most directly fulfills the project's Aim (Project Proposal, Section 1.2) and is rated **High** priority.

4.3.2 Stimulus/Response Sequences

- Tourists choose to add the currently viewed/listed attraction to their plan → system includes the attraction in the tourist's one-day visit plan.
- Tourist opens the itinerary planner view → system displays all attractions currently included in the plan.
- Tourists choose to remove an attraction from the plan → system removes it from the displayed itinerary.

4.3.3 Functional Requirements

- **FR-10:** The system shall allow the tourists to add an attraction to a personalized one-day visit plan.
- **FR-11:** The system shall allow the tourists to remove a previously added attraction from the one-day visit plan.

- **FR-12:** The system shall display the complete list of attractions currently included in the tourist one-day visit plan.
- **FR-13:** The system shall prevent the same attraction from being added to the one-day visit plan more than once.

Note: The Project Proposal explicitly excludes tourist user registration (Section 1.4.2). Accordingly, the one-day visit plan is scoped to the tourist's current browser session and does not persist to a user account across devices or sessions; this scoping is recorded as a system constraint (Section 2.5) rather than a separate requirement.

4.4 Administrator Authentication

4.4.1 Description and Priority

Restricts access to the attraction-management functions (Section 4.5) to authorized administrators only. Rated **High** priority, since it is the security control that protects the integrity of all attraction data relied upon by every Tourist User (NFR-04).

4.4.2 Stimulus/Response Sequences

- Administrator enters a username and password and submits the login form → system verifies the credentials and, if valid, grants access to the administrator module.
- Administrator submits invalid credentials → system denies access and displays an error message.
- Authenticated administrator chooses to log out → system ends the administrator session and returns to the login view.

4.4.3 Functional Requirements

- **FR-14:** The system shall require an administrator to log in with a valid username and password before accessing any attraction-management function.
- **FR-15:** The system shall deny access to the administrator module and display an error message when invalid login credentials are submitted.
- **FR-16:** The system shall allow an authenticated administrator to log out, ending the administrator session.

4.5 Attraction Record Management (Administrator Module)

4.5.1 Description and Priority

Allow an authenticated administrator to keep the attraction catalogue accurate and current by adding, editing, and removing attraction records. Rated **Medium-High** priority: used less frequently than the tourist-facing features, but essential to the long-term accuracy of the data those features depend on (Project Objective 5).

4.5.2 Stimulus/Response Sequences

- Authenticated administrator submits a new attraction record with all mandatory fields completed → system saves the new record and adds it to the catalogue.
- Authenticated administrator submits a new or edited record missing a mandatory field → system rejects the save and displays a validation message.

- Authenticated administrator edits an existing attraction record and saves the changes → system updates the stored record.
- Authenticated administrator chooses to delete an attraction record → system asks for confirmation before permanently removing it.

4.5.3 Functional Requirements

- **FR-17:** The system shall allow an authenticated administrator to add a new tourist attraction record, capturing name, category, description, opening hours, travel tips, approximate distance, images, and map location.
- **FR-18:** The system shall require the name, category, and description fields to be completed before a new or updated attraction record can be saved and shall display a validation message if any of these mandatory fields is missing.
- **FR-19:** The system shall allow an authenticated administrator to update the details of an existing tourist attraction record.
- **FR-20:** The system shall allow an authenticated administrator to delete an existing tourist attraction record.
- **FR-21:** The system shall require the administrator to confirm the action before an attraction record is permanently deleted.

5. Other Nonfunctional Requirements

The non-functional requirements below refine the eight quality-attribute categories identified in the Project Proposal (Section 1.11) into measurable, testable criteria, as required for this assignment. Where the Project Proposal used a qualitative target (e.g., “within a few seconds”), a specific, reasonable numeric target has been proposed here to make the requirement verifiable; these operationalized figures are indicated below and should be confirmed or adjusted during system testing.

5.1 Performance Requirements

- **NFR-01 (Performance):** The system shall display search results and attraction details within **3 seconds** of the tourist’s request, under normal operating conditions (broadband client connection, no more than 50 concurrent users), operationalizing the proposal’s stated goal of displaying results “within a few seconds” (Project Proposal, Section 1.11, NFR1).

5.2 Safety Requirements

The system is an informational planning tool and does not control physical equipment or processes; consequently, no requirements concerning physical loss, damage, or bodily harm apply in the traditional sense. The closest analogue to a “safety” concern is **information safety**: a tourist could be misled by inaccurate attraction data (e.g., incorrect opening hours). This risk is addressed through NFR-03 (Reliability, below) and through the data-verification assumption stated in Section 2.7 and Risk R1 of the Project Proposal (cross-verifying attraction data against official sources such as the Sri Lanka Tourism Development Authority before publication).

5.3 Security Requirements

- **NFR-04 (Security):** The system shall protect administrator access using a secure authentication mechanism (e.g., ASP.NET Core Identity with hashed and salted password storage), and shall reject any attempt to access an administrator function (FR-17 to FR-21) without a valid, authenticated administrator session, satisfying Project Proposal Section 1.11 (NFR4) and supporting Risk R6 mitigation (Section 1.13 of the proposal).
- The precise password-complexity policy and whether transport-level encryption (HTTPS/TLS) is mandated end-to-end are not specified in the Project Proposal and are recorded as [TBD] in Appendix C.

5.4 Software Quality Attributes

- **NFR-02 (Usability):** A first-time tourist user shall be able to locate and view the full details of a specific attraction in no more than **3 user interactions** (e.g., apply a search or filter, select an attraction, view its details), without needing external instructions, operationalizing the proposal’s usability goal (Section 1.11, NFR2).
- **NFR-03 (Reliability):** The system shall correctly process at least **99% of well-formed user requests** (search, filter, view details, itinerary actions) without an unhandled error during normal operation, operationalizing the proposal’s reliability goal of providing “accurate and consistent tourist information with minimal system errors” (Section 1.11, NFR3).

- **NFR-05 (Availability):** The system shall be accessible via a standard web browser and an active internet connection at any time, targeting at least **99% uptime**, excluding scheduled maintenance windows, consistent with Section 1.11 (NFR5) of the proposal.
- **NFR-06 (Maintainability):** The system shall allow an administrator to add, update, or delete an attraction record (FR-17 to FR-21) without requiring a system restart, a code change, or any interruption to the availability of other, unrelated attraction records, consistent with Section 1.11 (NFR6) of the proposal.
- **NFR-07 (Scalability):** The system's data model shall support the addition of new tourist attractions without requiring changes to existing application code, consistent with Section 1.11 (NFR7) of the proposal.
- **NFR-08 (Compatibility):** The system shall render and function correctly on the current major versions of Google Chrome, Microsoft Edge, and Mozilla Firefox, and shall provide a usable, responsively laid-out interface across screen widths from approximately 360 px (mobile) to 1920 px (desktop), consistent with Section 1.11 (NFR8) of the proposal.

5.5 Business Rules

- **BR-01:** Only an authenticated administrator may add, update, or delete a tourist attraction record (enforced by FR-14 to FR-21).
- **BR-02:** An attraction record may only be added to the catalogue if it is located within approximately 25 km of Kandy (Project Proposal, Sections 1.4.1–1.4.2).
- **BR-03:** Every attraction record must be assigned to exactly one of the seven predefined categories: Religious & Heritage, Nature, Adventure, Museum, Viewpoint, Recreation, or Eco Tourism (Project Proposal, Section 1.4.1).

6. Other Requirements

6.1 Database Requirements

Per the Project Proposal (Section 1.7), the SQL Server database, accessed through Entity Framework Core, must store at minimum the following conceptual entities:

- **Attraction:** name, category, description, opening hours, travel tips, approximate distance from Kandy, image(s), and map location (supports FR-01, FR-07, FR-17 to FR-20).
- **Category:** the seven predefined categories listed in BR-03, used to classify attractions and support filtering (FR-05).
- **Administrator Account:** credentials used for administrator authentication (FR-14).

Detailed schema design (tables, keys, and relationships) is a System Design-phase deliverable (Project Proposal, Section 1.12, “Database Design with ER Diagram,” Week 9 of the module schedule) and is outside the scope of this SRS; Appendix B provides a conceptual-level view only.

6.2 Requirements Traceability Matrix

The table below traces each user/stakeholder requirement (Section 2.7.1) forward to the functional and non-functional requirements that satisfy it, and identifies the corresponding item(s) in the Project Proposal’s own requirements table (Section 1.10/1.11) for backward traceability.

User/Stakeholder Requirement	Satisfied By	Project Proposal Source
UR-01 — View comprehensive attraction information	FR-01, FR-07	FR1, FR4
UR-02 — Search and filter by category	FR-02, FR-03, FR-04, FR-05, FR-06	FR2, FR3
UR-03 — View location/distance on a map	FR-08	FR5
UR-04 — Build a one-day itinerary	FR-10, FR-11, FR-12, FR-13	FR6
UR-05 — Secure administrator data management	FR-14, FR-15, FR-16, FR-17, FR-18, FR-19, FR-20, FR-21	FR7, FR8, FR9, FR10
UR-06 — Responsive, usable on any device	NFR-02, NFR-08	NFR2, NFR8

6.3 Internationalization

The Project Proposal does not specify a requirement for multilingual content or localization beyond English; no internationalization requirement is stated in this SRS. If multilingual support (e.g., Sinhala, Tamil) is later required, it would need to be added to the proposal scope before being reflected here.

Appendix A: Glossary

Term	Definition
API	Application Programming Interface.
Attraction	A tourist site or point of interest within approximately 25 km of Kandy that is catalogued by the system.
BA	Business Analysis / Business Analyst.
CSS	Cascading Style Sheets.
EF Core	Entity Framework Core — the ORM used to connect the ASP.NET Core Web API to SQL Server.
FR	Functional Requirement.
HTML	Hypertext Markup Language.
IEEE	Institute of Electrical and Electronics Engineers.
Itinerary / One-day visit plan	The personalized set of attractions a tourist selects for a single day's visit (FR-10 to FR-13).
JSON	JavaScript Object Notation.
NFR	Non-Functional Requirement.
ORM	Object-Relational Mapping.
REST	Representational State Transfer.
SDLC	Software Development Life Cycle.
SPA	Single Page Application.
SQL	Structured Query Language.
SRS	Software Requirements Specification (this document).
TBD	To Be Determined — an item not resolvable from the Project Proposal, listed in Appendix C.
UI	User Interface.
UNESCO	United Nations Educational, Scientific and Cultural Organization.
UR	User/Stakeholder Requirement.

Appendix B: Analysis Models

B.1 System Architecture (Layered View)

The system follows the three-tier layered architecture described in Section 2.1 (Presentation → Application/API → Data, with an external Google Maps service consumed by the Presentation layer). This is shown in Figure B-1 below (adapted from the Project Proposal, Section 1.7, Figure 1).

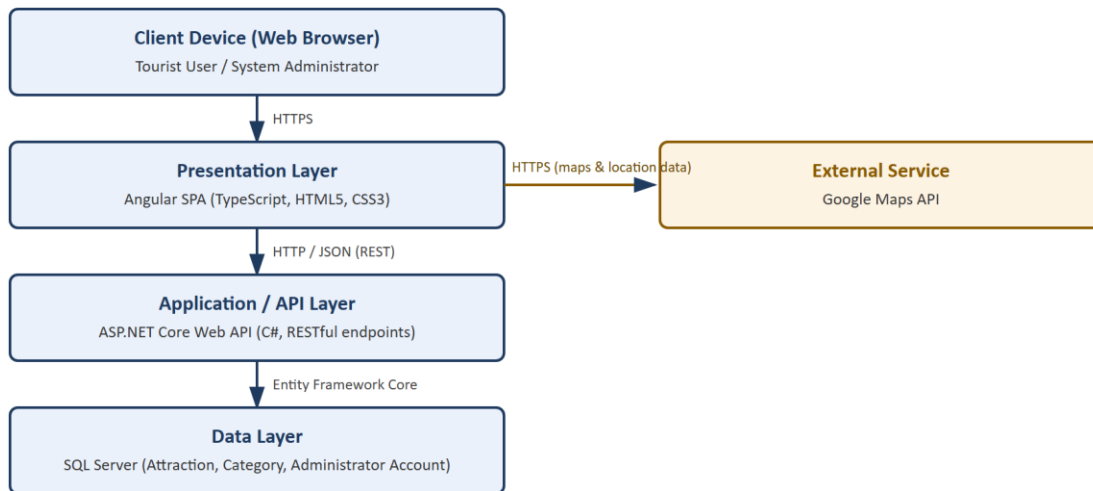


Figure B-1: System Architecture (Layered View) — three-tier architecture with an external Google Maps service consumed by the Presentation layer.

B.2 Use Case Diagram

The following use case diagram summarizes the system's two actors, Tourist User and System Administrator, and the primary use cases each performs. It provides a visual overview of the functional scope detailed in Section 4; the «include» relationship reflects that the administrator must log in (FR-14) before performing any attraction-management use case.

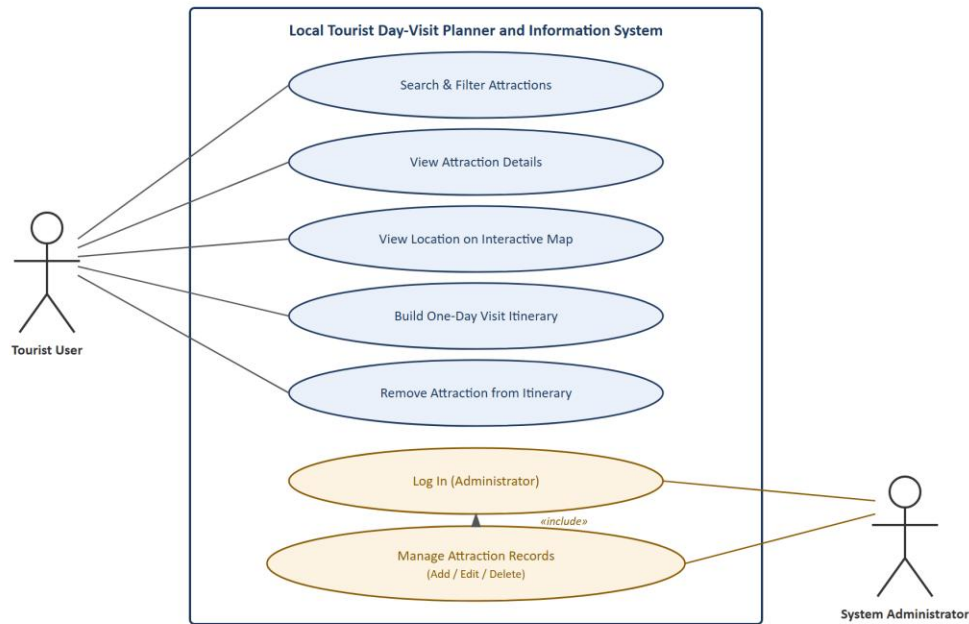


Figure B-2: Use Case Diagram — Tourist User and System Administrator interactions with the system.

B.3 Conceptual Data Model

The following conceptual entities and relationships are implied by the functional requirements in Section 4 and the database requirements in Section 6.1. This is a conceptual-level model only; the full Entity-Relationship Diagram is a System Design-phase deliverable (Project Proposal, Section 1.12). Figure B-3 below illustrates these entities and their attributes at the conceptual level.

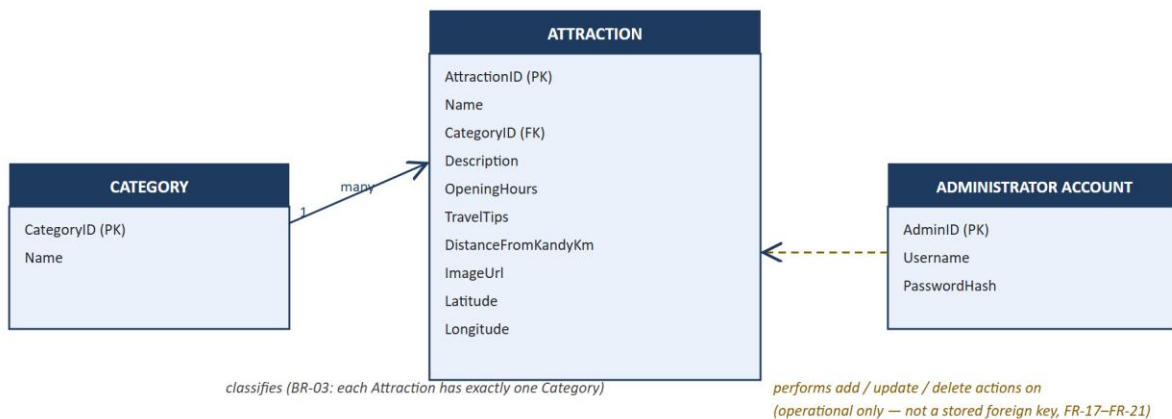


Figure B-3: Conceptual entity-relationship model (attribute-level detail is a System Design-phase deliverable).

- **Category (1) (many) Attraction:** each Attraction belongs to exactly one Category (BR-03); each Category may classify many Attractions.
- **Administrator Account:** independent entity used only for authentication (FR-14); has no stored relationship to Attraction records beyond the create/update/delete actions it is authorized to perform (FR-17 to FR-21).
- **One-day visit plan:** a session-scoped selection of Attraction records (FR-10 to FR-13); per the constraint noted in Section 4.3.3, this is not modelled as a persisted, per-user database entity, since tourist user registration is explicitly out of scope (Project Proposal, Section 1.4.2).

Appendix C: To Be Determined List

The following items could not be resolved from the Project Proposal and are recorded here, rather than being assumed, per the requirement that missing information be clearly identified. They should be resolved during the System Design phase (Weeks 7–8 of the module schedule).

TBD ID	Open Item Not Resolvable from the Project Proposal
TBD-1	The specific hosting/deployment environment for the completed system (cloud provider, server specification, or on premises) is not specified (Section 2.4).
TBD-2	Exact minimum supported browser version numbers for Chrome, Edge, and Firefox are not specified (Section 2.4, NFR-08).
TBD-3	No formal accessibility compliance level (e.g., WCAG 2.1 AA) is specified in the Project Proposal.
TBD-4	Whether HTTPS/TLS encryption is mandated end-to-end for all traffic, particularly administrator authentication traffic, is not explicitly stated (Section 3.4, Section 5.3).
TBD-5	The administrator password-complexity policy (minimum length, character requirements, expiry) is not specified (Section 5.3).
TBD-6	Formal user-documentation deliverables (user manual, administrator guide, in-app help) are not specified (Section 2.6).
TBD-7	Data backup, retention, and disaster-recovery procedures for the SQL Server database are not specified.
TBD-8	Whether the system will be deployed for use beyond the academic submission (i.e., a real production deployment for public use) is not specified.